

PROJECT DEVELOPMENT PHASE OF LEAVE MANAGEMENT SYSTEM

Development Environment

Platform Used

ServiceNow Developer Instance

Development Modules

- ServiceNow App Studio
- Flow Designer
- Business Rules
- Client Scripts
- Custom Tables
- Dashboards and Reports
- Form Designer
- Notifications

Application Development

Step 1: Creation of Application

A custom application named:

Leave Management System

was created using ServiceNow App Studio.

Purpose

The application serves as a centralized platform for managing employee leave requests, approvals, validations, notifications, and reporting.

Application Components

- Custom Table
- Leave Request Form
- Approval Flow

- Business Rule
- Client Script
- Dashboard
- Reports
- Notifications

4. Custom Table Development

A custom table was created to store employee leave request information.

Table Name

Leave Request

Fields Created

Field Name	Description
Employee Name	Employee requesting leave
Employee ID	Employee identifier
Leave Type	Type of leave requested
Start Date	Leave start date
End Date	Leave end date
Total Days	Total leave duration
Reason	Purpose of leave request
Approval Status	Approval state

Purpose

The table stores all leave request records and acts as the primary data repository for the application.

Benefits

- Centralized data storage
- Easy record retrieval
- Improved data consistency

5. Form Design Development

A user-friendly form was designed for employee leave request submission.

Form Fields

Employee Information

- Employee Name
- Employee ID

Leave Information

- Leave Type
- Start Date
- End Date
- Total Days
- Reason

Validation

Mandatory fields were configured to ensure complete leave request submission.

Benefits

- Improved usability
- Reduced data entry errors
- Standardized leave request collection
- Enhanced user experience

6. Leave Request Form Development

A custom leave request form was created to allow employees to submit leave requests.

Form Components

Field	Type
Employee Name	Single Line Text
Employee ID	Single Line Text
Leave Type	Choice
Start Date	Date
End Date	Date
Total Days	Integer
Reason	Multi-Line Text

Field	Type
Approval Status	Choice

Functionality

Employees can submit leave requests using a structured and user-friendly form. The form captures all necessary leave information and stores it in the Leave Request table.

7. Workflow Development Using Flow Designer

A workflow was created using Flow Designer to automate the leave request lifecycle.

Flow Name

Leave Approval Flow

Trigger

Leave Request Created

The workflow starts automatically whenever a new leave request record is created.

Initial Action

Update Status = Pending Approval

Purpose

The leave request is automatically marked as pending approval.

Approval Action

An approval step was configured using:

Ask For Approval

Purpose

The leave request is routed to the reporting manager for review and approval.

Approval Rule

Anyone Approves

Conditional Logic

An IF condition was configured.

Condition

Approval State = Approved

Purpose

The workflow proceeds only when approval is granted.

Status Update

If Approved

- Approval Status = Approved
- Notification Sent to Employee

If Rejected

- Approval Status = Rejected
- Notification Sent to Employee

Purpose

Automatically updates leave request status based on manager decision.

8. Business Rule Development

A Business Rule was configured to automatically calculate the total number of leave days.

Business Rule Name

Calculate Total Leave Days

Purpose

Automatically calculates leave duration based on the selected start date and end date.

Example

Start Date : 23-Jun-2026

End Date : 25-Jun-2026

Total Days : 3

Benefits

- Eliminates manual calculations
- Improves accuracy

- Reduces processing time
- Ensures consistency in leave records

9. Client Script Development

A Client Script was configured to validate leave dates before form submission.

Client Script Name

Leave Date Validation

Validation Rule

End Date cannot be earlier than Start Date.

Error Message

"End Date cannot be earlier than Start Date"

Purpose

Prevents invalid leave requests and improves data quality.

Benefits

- Reduces user errors
- Improves data accuracy
- Enhances form validation

10. Dashboard Development

A dashboard was created to monitor leave requests and approval status.

Dashboard Components

- Total Requests
- Approved Requests
- Rejected Requests
- Pending Requests
- Monthly Leave Statistics

Purpose

Provides real-time visibility into leave request activities and approval trends.

Benefits

- Quick monitoring
- Improved decision making
- Better leave management

11. Reporting Features

Reports were created to analyze leave request data.

Report Types

- Total Leave Requests Report
- Approved Requests Report
- Rejected Requests Report
- Pending Requests Report
- Monthly Leave Analysis Report

Benefits

- Improved monitoring
- Better decision making
- Enhanced reporting capabilities
- Easy performance analysis

12. Notification Management

Notifications were configured to keep users informed throughout the leave request process.

Notification Events

- Leave Request Submitted
- Leave Request Approved
- Leave Request Rejected
- Status Updated

Purpose

Provides timely communication and improves transparency between employees and managers.

Conclusion

The Project Development Phase successfully implemented the Leave Management System using ServiceNow. The application includes custom tables, forms, Flow Designer approvals, Business Rules for automatic leave calculation, Client Scripts for validation, notification management, and dashboards for reporting.

These features provide an efficient, automated, accurate, and user-friendly leave management solution that improves operational efficiency, reduces manual effort, and enhances employee experience.